

# Model Audit and Risk Management — handout

Lesson Forge Pro (owner)

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## Lesson plan

Model Audit and Risk Management

Bloom's levels: Evaluate, Create

Objectives

- Conduct model audits to identify and mitigate risks
- Develop risk management strategies to address model limitations
- Create model documentation to ensure transparency and reproducibility

Prerequisites

- Financial Modeling for Mergers and Acquisitions (M&A) and Investment Decisions

## Key points

Model Audit and Risk Management

Session Overview

In this 90-minute session, we will explore the importance of model audits and risk management in financial modeling. By the end of this session, you will be able to conduct model audits to identify and mitigate risks, develop risk management strategies to address model limitations, and create model documentation to ensure transparency and reproducibility.

Timed Agenda

| Time | Topic |

| --- | --- |

| 0:00 - 0:05 | Introduction and Objectives |

| 0:05 - 0:20 | Importance of Model Audits and Risk Management |

| 0:20 - 0:35 | Conducting Model Audits |

| 0:35 - 0:45 | Energizer/Break |

| 0:45 - 1:05 | Developing Risk Management Strategies |

| 1:05 - 1:20 | Creating Model Documentation |

| 1:20 - 1:30 | Conclusion and Next Steps |

Introduction and Objectives (0:00 - 0:05)

Welcome to the session on Model Audit and Risk Management. As financial modelers, we need to ensure that our models are accurate, reliable, and transparent. In this session, we will cover the following objectives:

Conduct model audits to identify and mitigate risks

Develop risk management strategies to address model limitations

Create model documentation to ensure transparency and reproducibility

Think of a model audit like a health check-up for your financial model. Just as a doctor checks your vital signs to ensure you're healthy, a model audit checks your model's vital signs to ensure it's accurate and reliable.

Importance of Model Audits and Risk Management (0:05 - 0:20)

Model audits and risk management are crucial in financial modeling because they help identify and mitigate potential risks. A model risk is like a ticking time bomb - it can explode at any moment, causing significant damage to your organization. By conducting regular model audits, you can detect potential risks and take corrective action before it's too late.

For example, imagine you're building a house. You need to ensure that the foundation is strong and stable before you start constructing the walls and roof. Similarly, in financial modeling, you need to ensure that your model is robust and reliable before you start making decisions based on its output.

Conducting Model Audits (0:20 - 0:35)

Conducting a model audit involves reviewing your model's assumptions, inputs, and outputs to ensure they are accurate and reliable. It's like checking the ingredients and recipe of your favorite dish to ensure it tastes good and is safe to eat.

Here are some steps to follow when conducting a model audit:

1. Review your model's documentation to ensure it's up-to-date and accurate.
2. Check your model's inputs to ensure they are reliable and consistent.
3. Verify your model's outputs to ensure they are accurate and reasonable.

Energizer/Break (0:35 - 0:45)

Let's take a short break to stretch our legs and refresh our minds. During this break, think about a time when you encountered a model risk in your work. How did you identify and mitigate it?

Developing Risk Management Strategies (0:45 - 1:05)

Developing risk management strategies involves identifying potential risks and taking corrective action to mitigate them. It's like having a fire evacuation plan in place - you hope you never need to use it, but it's essential to have one in case of an emergency.

Here are some steps to follow when developing risk management strategies:

1. Identify potential risks and prioritize them based on their likelihood and impact.
2. Develop corrective action plans to mitigate each risk.
3. Monitor and review your risk management strategies regularly to ensure they are effective.

For example, imagine you're building a bridge. You need to ensure that it can withstand strong winds and earthquakes. Similarly, in financial modeling, you need to ensure that your model can withstand potential risks and uncertainties.

Creating Model Documentation (1:05 - 1:20)

Creating model documentation is essential to ensure transparency and reproducibility. It's like writing a recipe for your favorite dish - you want to ensure that others can follow it and achieve the same result.

Here are some best practices to follow when creating model documentation:

1. Use clear and concise language to explain your model's assumptions and methodology.
2. Include diagrams and flowcharts to illustrate your model's logic and workflow.
3. Keep your documentation up-to-date and accurate.

Conclusion and Next Steps (1:20 - 1:30)

In conclusion, model audits and risk management are essential in financial modeling. By conducting regular model audits, developing risk management strategies, and creating model documentation, you can ensure that your models are accurate, reliable, and transparent.

As a next step, I encourage you to apply the concepts learned in this session to your own work.

Remember, a model audit is like a health check-up for your financial model - it's essential to ensure it's healthy and reliable.

Anticipated Learner Questions and Answers

Q: What is the difference between a model audit and risk management?

A: A model audit is a review of your model's assumptions, inputs, and outputs to ensure they are accurate and reliable. Risk management involves identifying potential risks and taking corrective action to mitigate them.

Q: How often should I conduct a model audit?

A: It depends on the complexity and criticality of your model. As a general rule, you should conduct a model audit at least once a year, or whenever you make significant changes to your model.

Q: What are some common model risks that I should be aware of?

A: Some common model risks include data errors, assumption errors, and methodology errors. You should also be aware of external risks such as changes in market conditions or regulatory requirements.